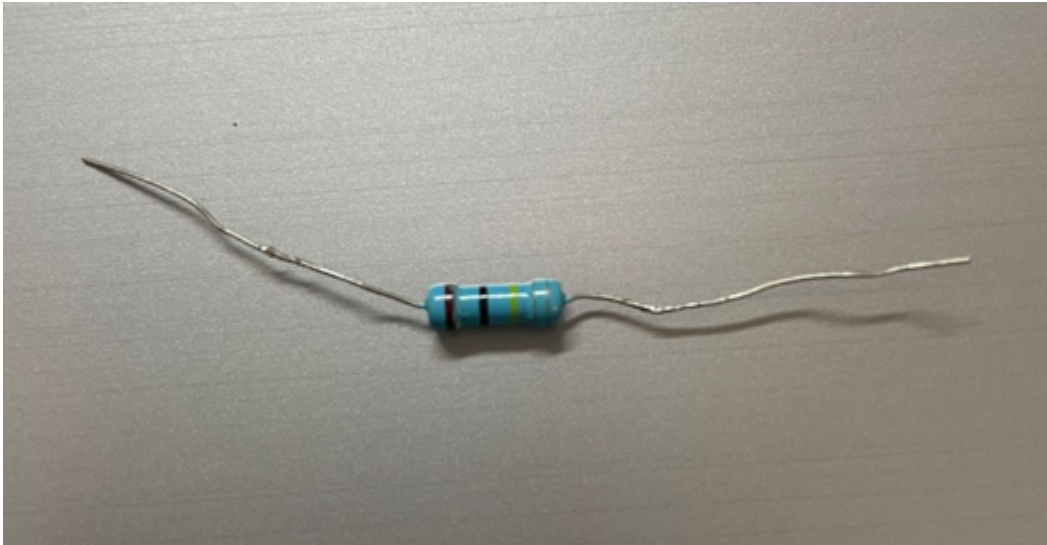


GROUP 2: BINARY FORCE

Experiment 1:

1. Pick one resistor, take a picture of it and determine its resistance value based on the colour code. Write down the resistance value.



Resistance value: $100\text{ k}\Omega = 10,000\ \Omega$

2. Using a multimeter, measure the resistance value emitted by the resistor. Take a picture of this setup and write down the measured resistance.

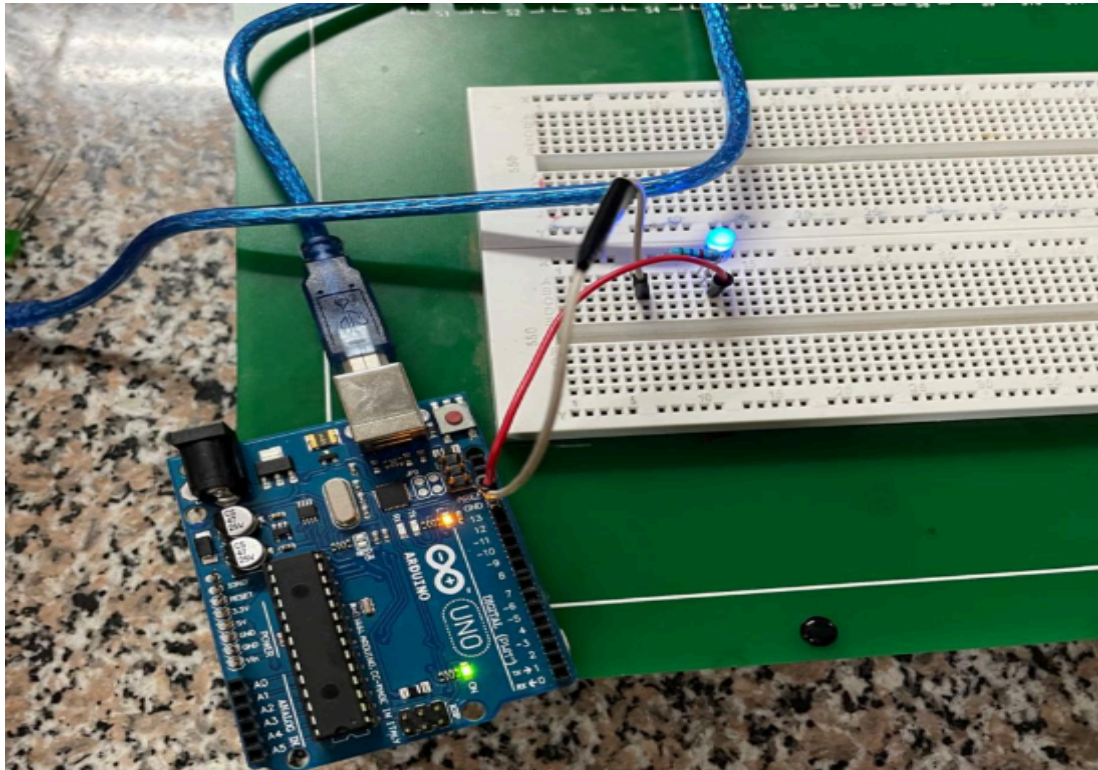


Measured resistance: $97.1\text{ k}\Omega = 97,100\ \Omega$

Experiment Reflection: Where does the difference in the resistance values arise from? The difference happens because factory-made resistors have an allowed error margin (tolerance of + or - 5% since the band is gold), and the multimeter itself may be having minor reading inaccuracies.

Experiment 2:

1. Using the breadboard, microcontroller, and LED, recreate the following schematic. Take a picture of the physical implementation.



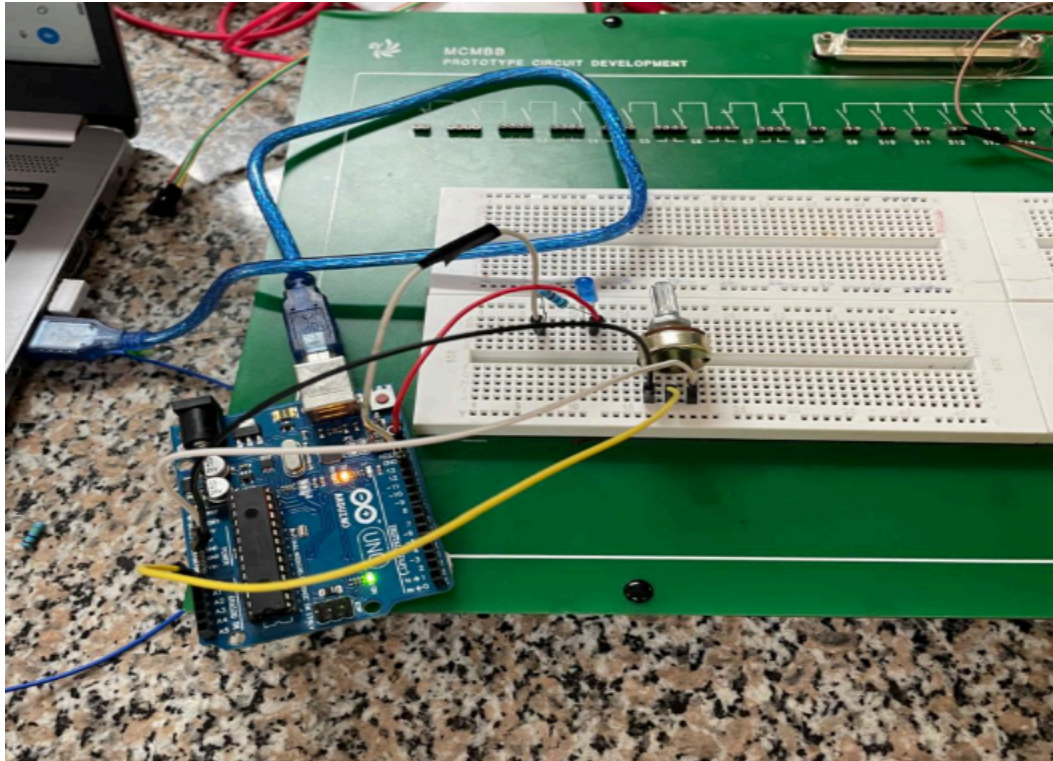
2. In Arduino IDE, deploy the Blink program which is usually inbuilt in Arduino and upload to your board. Congratulations you just created your first sketch! :)

3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered.

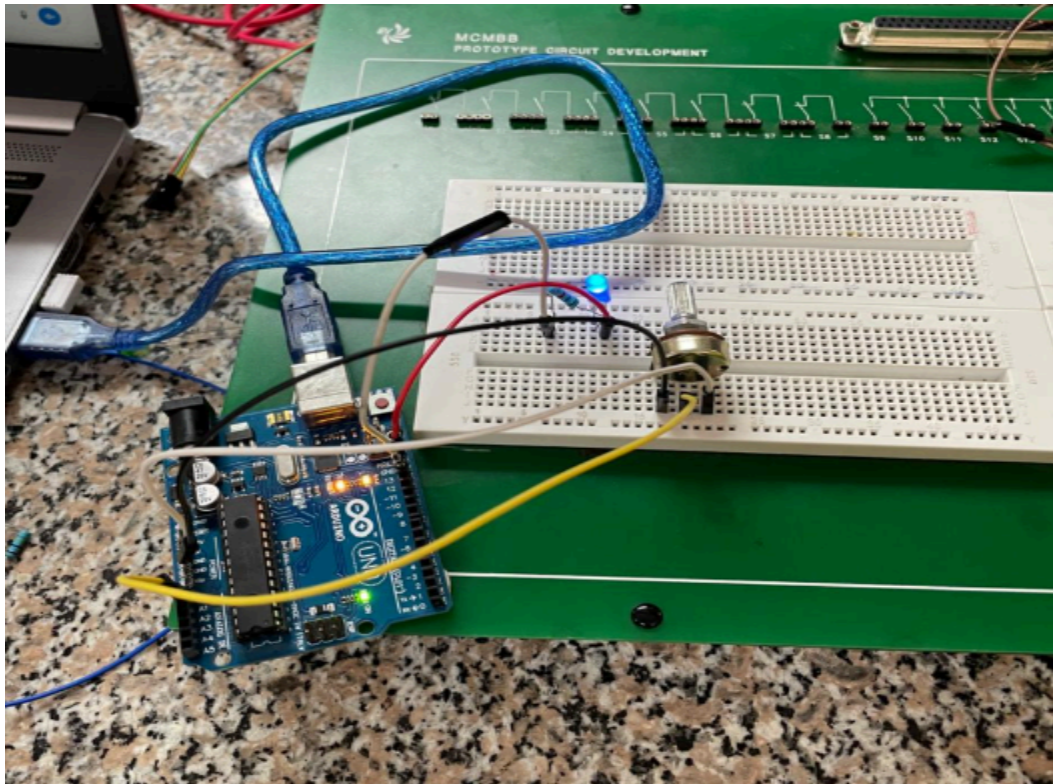
Voltage: 2.536V

4. Reconfigure the circuit you created in question 1 by adding a potentiometer to simulate Pulse Width Modulation (PWM) on the LED. Take 3 images of your circuit with the LED at 25% brightness, 50% brightness and 100% brightness.

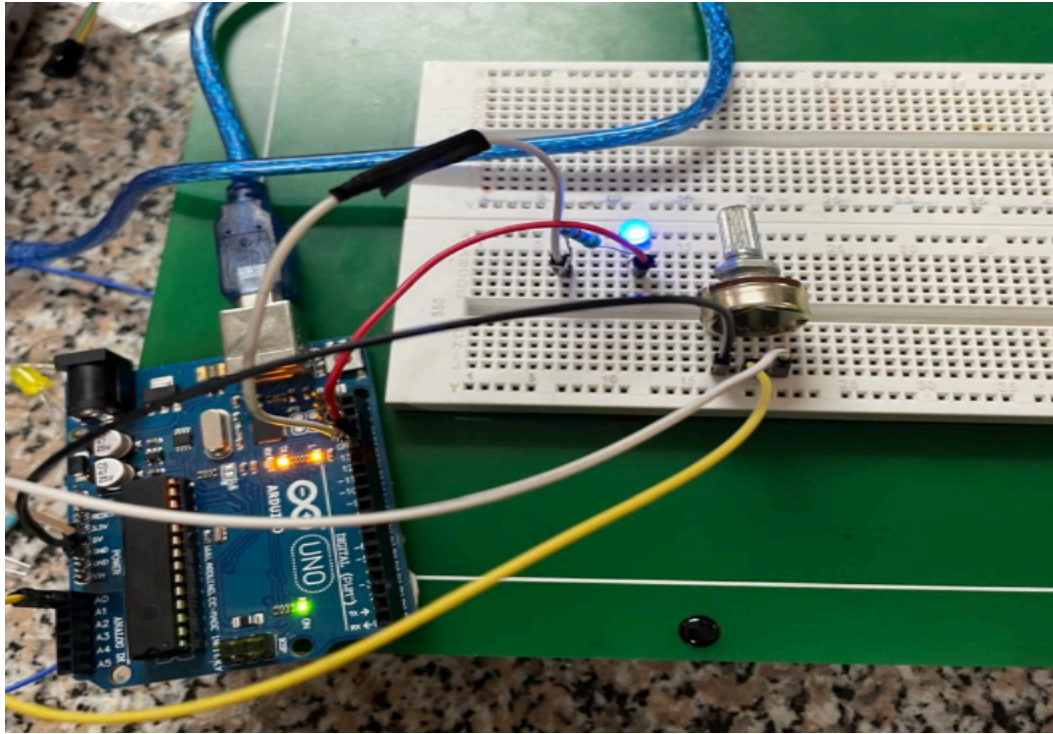
25% brightness



50% Brightness

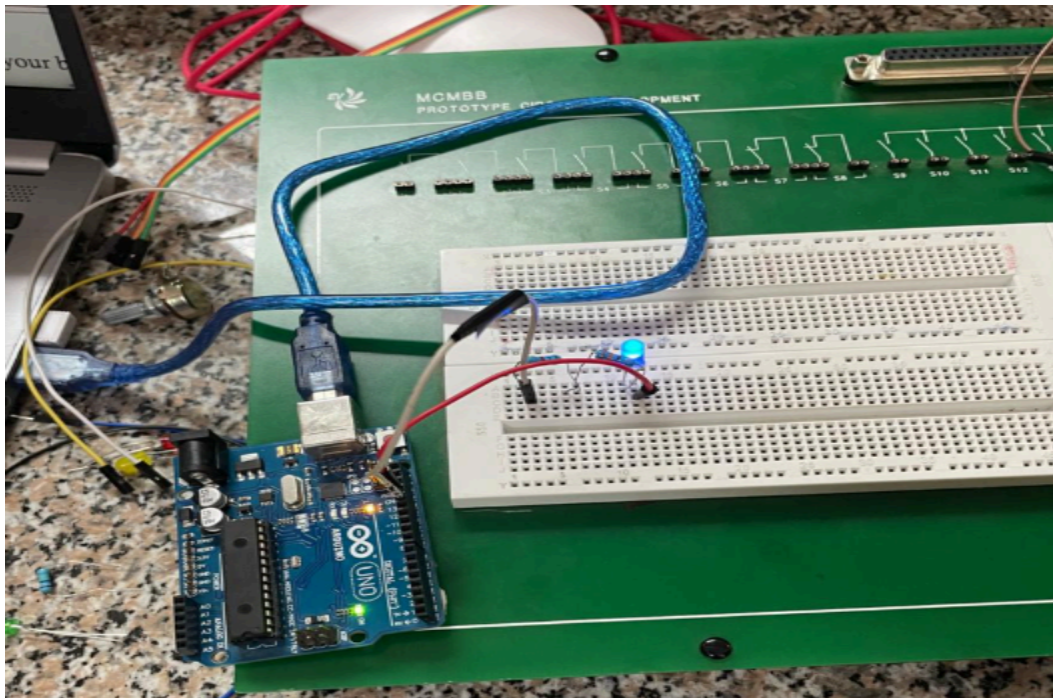


100% Brightness



Experiment 3:

1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in series. Take a picture of the physical implementation.



2. In Arduino IDE, re-deploy the Blink program and upload to your board.

3. Thereafter, use the multimeter to determine the voltage passing through the circuit.
Write down the value registered at each resistor.

Voltage: 1.527V

Experiment Reflection: Use Ohm's Law to estimate the current passed at each resistor.

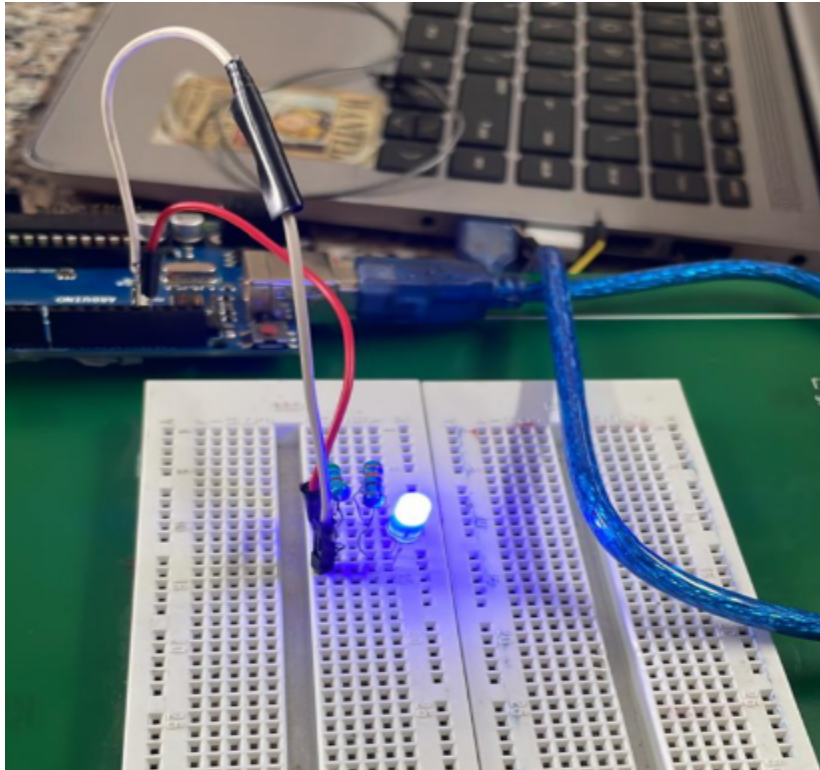
Current = Voltage / Resistance

Current = 1.527V / 97, 100 Ω

= 0.0001573 A = 15.73 μ A

Experiment 4:

1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in parallel. Take a picture of the physical implementation.



2. In Arduino IDE, re-deploy the Blink program and upload to your board.

3. Thereafter, use the multimeter to determine the voltage passing through the circuit.
Write down the value registered at each resistor.

First Resistor: 2.53 V

Second Resistor: 2.79 V

Experiment Reflection: Use Ohm's Law to estimate the current passed at each resistor

Resistor 1:

Current = $2.53 \text{ V} / 97,100 \Omega = 0.00002606 \text{ A} = 26.06 \mu\text{A}$

Resistor 2:

Current = $2.79 \text{ V} / 97,100 \Omega = 0.00002873 \text{ A} = 28.73 \mu\text{A}$

GROUP PICTURE

